# TP : OpenSSL

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## Mission 1

Mission 1 :

1) Public key : $n = pq = 11 \times 5 = 55$ $(55, 3)$

Private key :

$\varphi(n) = (p-1)(q-1) = 40$

$ed \equiv 1 \ [40]$ $e = 3$

$d = \frac{81}{3} = 27$

2) $m = C^d \bmod (n)$

$= 23^{27} \bmod (55)$

$23^{27} = (23^4)^6 \, 23^2 \, 23$

$23^4 = 1 \bmod [55]$
$23^2 = 34 \bmod [55]$
$23 = 23 \bmod [55]$

$m = 23^{27} \bmod [55]$

$= (1)^6 \times 34 \times 23 \ [55]$

$= 12 \ [55]$

$m = 12$

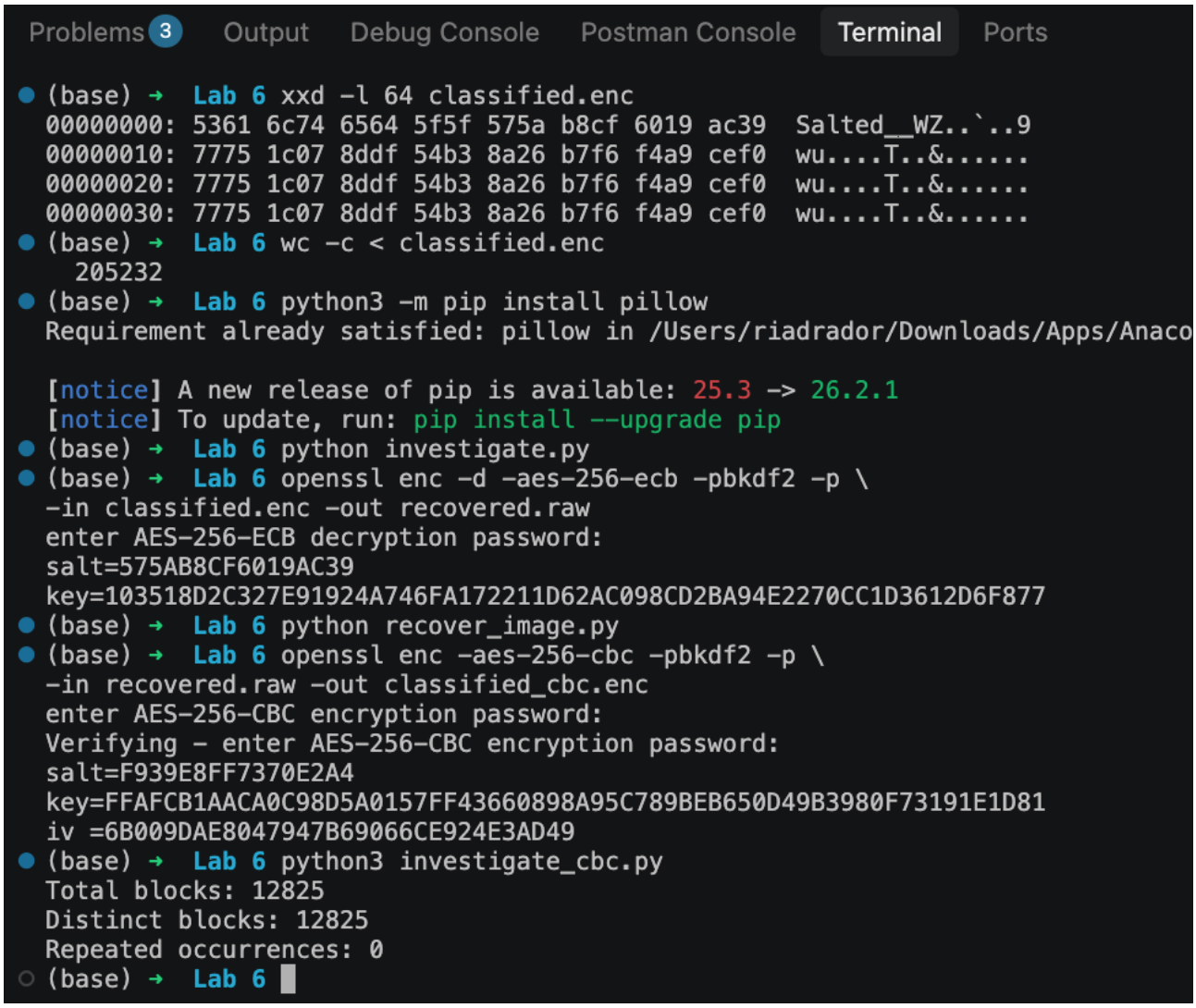
3) To verify

$C = m^e \bmod (n)$

$C = 12^3 \bmod (55)$

$C = 23$

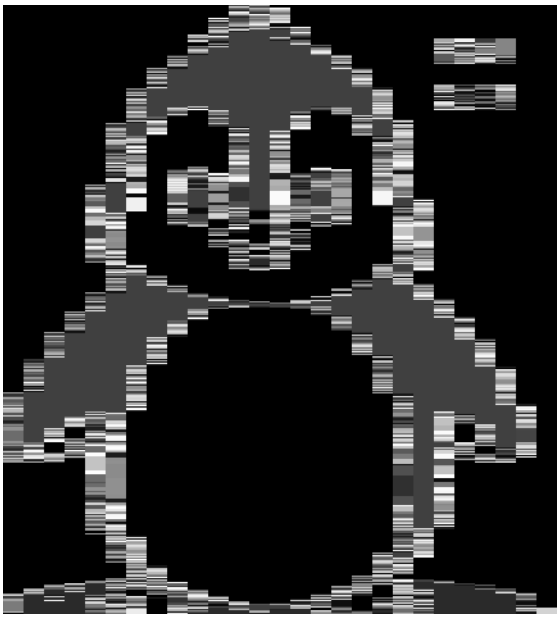
## Mission 2




The first 8 bytes contain `Salted__`; the next 8 bytes contain the salt `575AB8CF6019AC39`. Exclude the first 16 bytes before analyzing the ciphertext.

For the visualization, retain the next $432 \times 475 = 205200$ bytes, excluding the final 16-byte padding block.

Please refer to `investigate.py` in the GitHub repo.



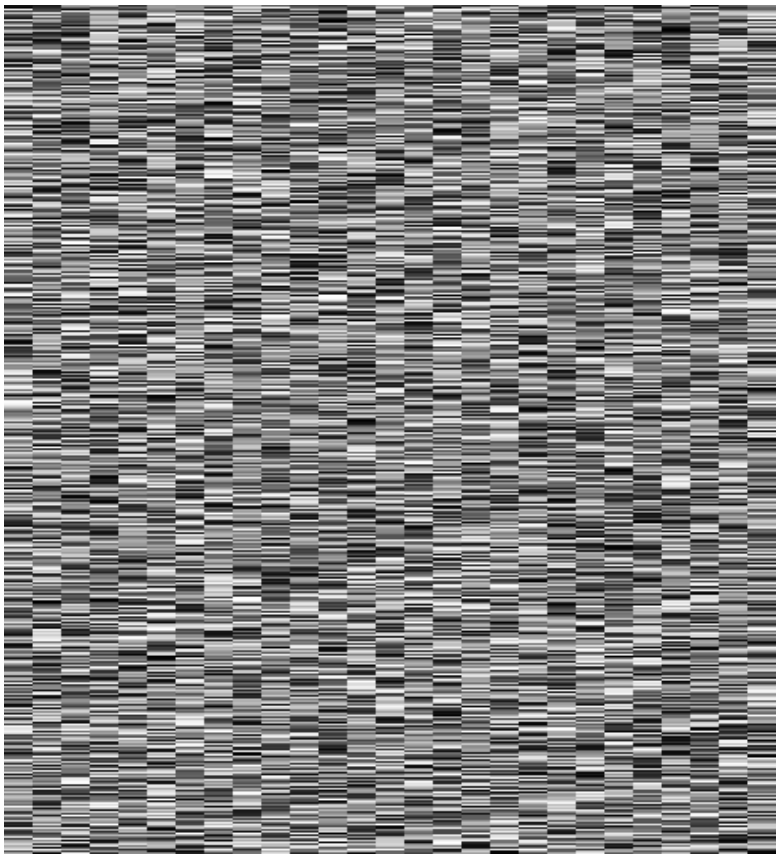
A penguin silhouette is recognizable through repeated ECB blocks.

Please refer to `recover_image.py` in the GitHub repo.




Decryption using password `12` reveals the original penguin and the message "YOU WIN".

Please refer to `investigate_cbc.py` in the GitHub repo.



The original picture is no longer recognizable.